HAN LI

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Database Systems · Distributed Systems · Concurrency Control

EDUCATION

University at Buffalo, SUNY Sep 2024 - Present

Ph.D. in Computer Science and Engineering · Advisor: Prof. Haonan Lu · Expected graduation: 2028

Tongji University Sep 2015 - Jul 2019

B.Eng. in Software Engineering

RESEARCH EXPERIENCE

Juicer: Proactive Ordering & Conflict Pacing Sep 2024 - Present

Research Software Engineer · Planned submission: OSDI 2027

- Characterizing abort causes across concurrency-control protocols: cross-shard order inversions and conflicts involving requests, control messages, and transaction semantics. Proposing a **horizontal approach** that enhances existing protocols through proactive ordering and conflict pacing.
- Prototyping **RPC-layer gRPC interceptors** for easy adoption across systems while retaining existing correctness mechanisms; evaluating CockroachDB and Spanner/NCC research implementations.

Lakehouse Concurrency Control Apr 2026 - Present

Research Software Engineer · Planned submission: OSDI 2027

- Investigating how commit conflicts trigger **full-statement retries**, wasting completed computation and S3 requests. Exploring early read/write intents and **incremental repair** to preserve unaffected outputs and make repair proportional to affected data.
- Prototyping Spark/Iceberg coordination and repair for restricted MERGE statements, validating against committed snapshots; using concurrent workloads and S3 tracing to assess correctness and wasted work.

INDUSTRY EXPERIENCE

Alibaba Group · Dataphin Aug 2019 - Jul 2022

Software Development Engineer II · Earlier internship: Jul 2018 - Jan 2019

- Built scheduling for **TB-scale memory resources**: implemented a Mesos framework and Rs-Agent's asynchronous event loop and task state machine for buffering and local execution.
- Evaluated worker designs in YARN and Mesos and batch scheduling in KubeBatch and Volcano; designed a subsequent **Kubernetes-based scheduling architecture**.
- Led a **10-person cross-functional team** on a customer-tagging platform; coordinated delivery with product and customers, compiled tag configurations into Hive/MaxCompute jobs, and resolved execution dependencies.
- Built behavior-cleansing, tag-computation, cumulative-tag and statistics pipelines with Hive SQL and Java UDFs. Developed user-keyed wide tables and a **materialized-view abstraction** supporting column pruning and time-based optimization, alongside tag-coverage reporting and data exploration.
- Built Hive metadata discovery and RocketMQ-based change delivery, indexed with **Elasticsearch/PostgreSQL GIN**; provided unified discovery and search for tables, columns, functions and workspaces.

TECHNICAL SKILLS

Languages: Java, Go, C++, Python, SQL
Data systems: Spark, Iceberg, Hive, Flink
Infrastructure: Kubernetes, AWS, Prometheus
Coding Agent: Claude Code, DeepSeek Harness

Review draft